TABLE STRUCTURE - PART A

PART A

1. **Teaching Process**

   Lectures, Tutorials, Practical's, Mini/Major project etc.

Column	Meaning	Java Type	DB Type
SN	Serial number	`Integer`	`INT`
Semester	Semester number/text	`Integer` *(or String if "Sem 1")*	`INT` / `VARCHAR`
Course Code / Name	Course identifier	`String`	`VARCHAR(255)`
No. of Classes (planned)	Total expected classes	`Integer`	`INT`
No. of Classes (conducted)	Actual classes taken	`Integer`	`INT`
API Score (Faculty)	Score given by faculty	`Double`	`DOUBLE`
API Score (HOD)	Score by HOD	`Double`	`DOUBLE`
Signature	Approval / sign flag	`String` *(or Boolean)*	`VARCHAR(100)` / `BOOLEAN`

🎯 Access Control for Your APIs

✅ 1. CREATE (POST)

`POST /api/teaching-process`

- 👤 **FACULTY → ✅ Allowed**
- 🛠️ **ADMIN → ❌ Not needed**

👉 Faculty fills their own teaching data

✅ 2. READ (GET)

`GET /api/teaching-process/faculty/{facultyId}`

- 👤 **FACULTY → ✅ Only their own data**
- 🛠️ **ADMIN → ✅ Can view anyone's data**

`GET /api/teaching-process`

- 👤 **FACULTY → ❌**
- 🛠️ **ADMIN → ✅ Full access**

👉 Admin dashboard view

✅ 3. UPDATE (PUT)

`PUT /api/teaching-process/{id}`

Split logic based on role:

👤 FACULTY can update:

- plannedClasses
- conductedClasses
- courseCodeName

❌ Cannot update:

- apiScoreHod
- signature

🛠️ ADMIN can update:

- apiScoreHod
- signature
- (optionally everything)

✅ 4. DELETE

`DELETE /api/teaching-process/{id}`

- 👤 **FACULTY** → ❌
- 🛠️ **ADMIN** → ✅

👉 In real systems → use **soft delete** instead

Course File table

Column	Meaning	Java Type	DB Type
SN	Serial number	`Integer`	`INT`
Course / Paper	Course name/code	`String`	`VARCHAR(255)`
Title	Topic/title of file	`String`	`VARCHAR(255)`
Details (Yes/No with proofs)	Whether proof exists	`Boolean` *(better)*	`BOOLEAN`
API Score (Faculty)	Faculty score	`Double`	`DOUBLE`
API Score (HOD)	HOD score	`Double`	`DOUBLE`
Signature	Approval status	`String` *(or Boolean)*	`VARCHAR(100)`

API Endpoints

🔹 Base URL

`/api/course-files`

✅ CREATE (POST)

`POST /api/course-files`

👤 Faculty

- ✅ Can create entry
- Fields they fill:
  - course/paper
  - title

- details (proof yes/no)

🛠️ Admin

- ❌ Not needed

✅ READ (GET)

Get by Faculty (Main API)

`GET /api/course-files/faculty/{facultyId}`

- 👤 Faculty → only their data
- 🛠️ Admin → any faculty

Get All

`GET /api/course-files`

- 👤 Faculty → ❌
- 🛠️ Admin → ✅

✅ UPDATE (PUT)

`PUT /api/course-files/{id}`

👤 Faculty can update:

- course/paper
- title
- details (proof)

❌ Cannot update:

- apiScoreHod
- signature

🛠️ Admin can update:

- apiScoreHod
- signature
- (optionally all fields)

✅ DELETE

`DELETE /api/course-files/{id}`

- 👤 Faculty → ❌
- 🛠️ Admin → ✅

Use of participatory and innovative teaching-learning methodologies; updating of subject content, course improvement etc.

Column	Meaning	Java Type	DB Type
SN	Serial number	`Integer`	`INT`
Short Description	Activity name	`String`	`VARCHAR(255)`
Details (Yes/No with proofs)	Whether implemented	`Boolean`	`BOOLEAN`
API Score (Faculty)	Score by faculty	`Double`	`DOUBLE`
API Score (HOD)	Score by HOD	`Double`	`DOUBLE`
Signature	Approval	`String` *(or Boolean)*	`VARCHAR(100)`

API Endpoints

🔹 Base URL

`/api/teaching-methods`

✅ CREATE (POST)

`POST /api/teaching-methods`

👤 Faculty

- ✅ Can mark activity as done (true/false)
- Can add proof later

🛠️ Admin

- ❌ Not needed

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/teaching-methods/faculty/{facultyId}`

- 👤 Faculty → own data
- 🛠️ Admin → any faculty

Get All

`GET /api/teaching-methods`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/teaching-methods/{id}`

👤 Faculty can update:

- details (true/false)
- proof (if added later)

❌ Cannot update:

- apiScoreHod
- signature

🛠️ Admin can update:

- apiScoreHod
- signature

✅ DELETE

`DELETE /api/teaching-methods/{id}`

- 👤 ❌
- 🛠️ ✅

Project

Column	Meaning	Java Type	DB Type
SN	Serial number	`Integer`	`INT`
Project	Type of project activity	`String` *(or ENUM)*	`VARCHAR(255)`
API Score (Faculty)	Score by faculty	`Double`	`DOUBLE`
API Score (HOD)	Score by HOD	`Double`	`DOUBLE`
API Score (Director)	Score by Director	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/projects`

✅ CREATE (POST)

`POST /api/projects`

👤 Faculty

- ✅ Can create/update their project entries

👉 But better approach:

- Pre-create 4 rows per faculty

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/projects/faculty/{facultyId}`

- 👤 Faculty → own data
- 🛠️ Admin → any data

Get All

`GET /api/projects`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/projects/{id}`

👤 Faculty can update:

- their own contribution data (if any future fields added)

⚠️ Usually faculty gives initial score

🛠️ HOD can update:

- apiScoreHod

🛠️ Director can update:

- apiScoreDirector

🔥 Role Breakdown

Field	Faculty	HOD	Director
apiScoreFaculty	✅	❌	❌
apiScoreHod	❌	✅	❌
apiScoreDirector	❌	❌	✅

✅ DELETE

`DELETE /api/projects/{id}`

- 👤 ❌
- 🛠️ Admin → ✅ (rarely used)

Qualification Enhancement

Column	Meaning	Java Type	DB Type
SN	Serial number	`Integer`	`INT`
Project	Type of qualification	`ENUM / String`	`VARCHAR(100)`
API Score (Faculty)	Score by faculty	`Double`	`DOUBLE`
API Score (HOD)	Score by HOD	`Double`	`DOUBLE`
API Score (Director)	Score by Director	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/qualification-enhancement`

✅ CREATE (POST)

`POST /api/qualification-enhancement`

- 👤 Faculty → ✅
- 🛠️ Admin → ❌

👉 Better: pre-create 2 rows per faculty

✅ READ (GET)

Get by Faculty

`GET /api/qualification-enhancement/faculty/{facultyId}`

- 👤 own data
- 🛠️ any data

Get All

`GET /api/qualification-enhancement`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/qualification-enhancement/{id}`

👤 Faculty:

- can update `apiScoreFaculty`

🛠️ HOD:

- can update `apiScoreHod`

🛠️ Director:

- can update `apiScoreDirector`

✅ DELETE

`DELETE /api/qualification-enhancement/{id}`

- 👤 ❌
- 🛠️ Admin → ✅

🧠 Role Control Summary

Field	Faculty	HOD	Director
apiScoreFaculty	✅	❌	❌
apiScoreHod	❌	✅	❌
apiScoreDirector	❌	❌	✅

For qualification table 2

2. Teaching Process Summary (VERY IMPORTANT)

This part:

```
Total marks out of 100 (i+ii+iii+iv+v)
Marks scaled to out of 25
```

❗ DO NOT CREATE A TABLE FOR THIS

👉 This is **computed data**, not stored data

✅ How to Handle

- Calculate dynamically:
  - Sum of all Teaching Process sections
  - Scale to 25

🔥 API for This

Endpoint:

`GET /api/teaching-process/summary/{facultyId}`

Returns:

- totalMarksOutOf100
- scaledMarksOutOf25

B: Students' Feedback

Column	Meaning	Java Type	DB Type
S. No	Serial number	`Integer`	`INT`
Course Code / Name	Course identifier	`String`	`VARCHAR(255)`
First Feedback	Rating (out of 5)	`Double`	`DOUBLE`
Second Feedback	Rating (out of 5)	`Double`	`DOUBLE`
Average	Avg of above two	`Double` *(computed)*	❌ Do NOT store
API Score (Faculty)	Score	`Double`	`DOUBLE`
API Score (HOD)	Score	`Double`	`DOUBLE`
API Score (Director)	Score	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/student-feedback`

✅ CREATE (POST)

`POST /api/student-feedback`

👤 Faculty

- ✅ Can add:
  - course
  - firstFeedback
  - secondFeedback

🛠️ Admin

- ❌ Not needed

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/student-feedback/faculty/{facultyId}`

- 👤 own data
- 🛠️ any data

👉 Response should include:

- calculated `average`

Get All

`GET /api/student-feedback`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/student-feedback/{id}`

👤 Faculty can update:

- firstFeedback
- secondFeedback
- course

❌ Cannot update:

- apiScoreHod
- apiScoreDirector

🛠️ HOD:

- can update `apiScoreHod`

🛠️ Director:

- can update `apiScoreDirector`

✅ DELETE

`DELETE /api/student-feedback/{id}`

- 👤❌
- 🛠️ Admin → ✅

🧠 Additional API (VERY IMPORTANT)

📊 Get Total Average (Bottom Row)

`GET /api/student-feedback/summary/{facultyId}`

👉 Returns:

- overallAverage (across all subjects)

C: **Departmental / School Activities**

Column	Meaning	Java Type	DB Type
S. No	Serial number	`Integer`	`INT`
Activity	Name of activity	`String`	`VARCHAR(255)`
Nature of activity	Description/type	`String`	`TEXT`
API Score (Faculty)	Score	`Double`	`DOUBLE`
API Score (HOD)	Score	`Double`	`DOUBLE`
API Score (Director)	Score	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/department-activities`

✅ CREATE (POST)

`POST /api/department-activities`

👤 Faculty

- ✅ Can add:
  - activity
  - nature of activity

🛠️ Admin

- ❌ Not needed

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/department-activities/faculty/{facultyId}`

- 👤 own data
- 🛠️ any data

Get All

`GET /api/department-activities`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/department-activities/{id}`

👤 Faculty can update:

- activity
- nature of activity
- apiScoreFaculty

❌ Cannot update:

- apiScoreHod
- apiScoreDirector

🛠️ HOD:

- can update `apiScoreHod`

🛠️ Director:

- can update `apiScoreDirector`

✅ DELETE

`DELETE /api/department-activities/{id}`

- 👤 ❌
- 🛠️ Admin → ✅

🧠 Additional API (IMPORTANT)

📊 Total Score

`GET /api/department-activities/summary/{facultyId}`

👉 Returns:

- totalScore (max 20)

D: University Level Activities

Column	Meaning	Java Type	DB Type
Sr. No	Serial number	`Integer`	`INT`
Activity	Name of activity	`String`	`VARCHAR(255)`
Nature of activity	Description/type	`String`	`TEXT`
API Score (Faculty)	Score	`Double`	`DOUBLE`
API Score (HOD)	Score	`Double`	`DOUBLE`
API Score (Director)	Score	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/university-activities`

✅ CREATE (POST)

`POST /api/university-activities`

👤 Faculty

- ✅ Can add:
  - activity
  - nature of activity

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/university-activities/faculty/{facultyId}`

- 👤 own data
- 🛠️ admin → any data

Get All

`GET /api/university-activities`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/university-activities/{id}`

👤 Faculty can update:

- activity
- nature of activity
- apiScoreFaculty

❌ Cannot update:

- apiScoreHod
- apiScoreDirector

🛠️ HOD:

- can update `apiScoreHod`

🛠️ Director:

- can update `apiScoreDirector`

✅ DELETE

`DELETE /api/university-activities/{id}`

- 👤 ❌
- 🛠️ Admin → ✅

🧠 Additional API (IMPORTANT)

📊 Total Score

`GET /api/university-activities/summary/{facultyId}`

👉 Returns:

- totalScore (max 30)

E: Contribution to Society

Column	Meaning	Java Type	DB Type
Sr. No	Serial number	`Integer`	`INT`
Activity	Type of social activity	`ENUM / String`	`VARCHAR(100)`
Details of activity	Description / proof	`String`	`TEXT`
API Score (Faculty)	Score	`Double`	`DOUBLE`
API Score (HOD)	Score	`Double`	`DOUBLE`
API Score (Director)	Score	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/social-contributions`

✅ CREATE (POST)

`POST /api/social-contributions`

👤 Faculty

- ✅ Can:
  - select activity type
  - add details

👉 Better:

- Pre-create 9 rows per faculty

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/social-contributions/faculty/{facultyId}`

- 👤 own data
- 🛠️ admin → any data

Get All

`GET /api/social-contributions`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/social-contributions/{id}`

👤 Faculty can update:

- detailsOfActivity
- apiScoreFaculty

❌ Cannot update:

- apiScoreHod
- apiScoreDirector

🛠️ HOD:

- can update `apiScoreHod`

🛠️ Director:

- can update `apiScoreDirector`

✅ DELETE

`DELETE /api/social-contributions/{id}`

- 👤❌
- 🛠️ Admin → ✅

🧠 Additional API (IMPORTANT)

📊 Total Score

`GET /api/social-contributions/summary/{facultyId}`

👉 Returns:

- totalScore (max 10)

F: Industry Connect Activity

Column	Meaning	Java Type	DB Type
Sr. No	Serial number	`Integer`	`INT`
Name of Industry	Industry/company name	`String`	`VARCHAR(255)`
Details of activity	Description	`String`	`TEXT`
API Score (Faculty)	Score	`Double`	`DOUBLE`
API Score (HOD)	Score	`Double`	`DOUBLE`
API Score (Director)	Score	`Double`	`DOUBLE`

API Endpoints

🔹 Base URL

`/api/industry-connect`

✅ CREATE (POST)

`POST /api/industry-connect`

👤 Faculty

- ✅ Can add:
  - industry name
  - activity details

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/industry-connect/faculty/{facultyId}`

- 👤 own data
- 🛠️ admin → any data

Get All

`GET /api/industry-connect`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/industry-connect/{id}`

👤 Faculty can update:

- industry name
- details
- apiScoreFaculty

❌ Cannot update:

- apiScoreHod
- apiScoreDirector

🛠️ HOD:

- can update `apiScoreHod`

🛠️ Director:

- can update `apiScoreDirector`

✅ DELETE

`DELETE /api/industry-connect/{id}`

- 👤 ❌
- 🛠️ Admin → ✅

🧠 Additional API

📊 Total Score

`GET /api/industry-connect/summary/{facultyId}`

👉 Returns:

- totalScore (max 5)

G: Annual Confidential Report (ACR)

Column	Meaning	Java Type	DB Type
S. No	Serial number	`Integer`	`INT`
Subject	Evaluation criteria	`ENUM / String`	`VARCHAR(100)`
API Score (HOD)	Score by HOD	`Double`	`DOUBLE`
API Score (Director)	Score by Director	`Double`	`DOUBLE`
Signature	Approval status	`String` *(or Boolean)*	`VARCHAR(100)`

API Endpoints

🔹 Base URL

`/api/acr`

✅ CREATE (POST)

`POST /api/acr`

👉 Only used for:

- Pre-creating 5 rows per faculty

👤 Faculty → ❌

🛠️ Admin → ✅ (system setup)

✅ READ (GET)

Get by Faculty (MAIN)

`GET /api/acr/faculty/{facultyId}`

- 👤 Faculty → ✅ (view only)
- 🛠️ Admin → ✅

Get All

`GET /api/acr`

- 👤 ❌
- 🛠️ ✅

✅ UPDATE (PUT)

`PUT /api/acr/{id}`

👤 Faculty

- ❌ NO ACCESS

🛠️ HOD can update:

- apiScoreHod

🛠️ Director can update:

- apiScoreDirector
- signature

✅ DELETE

`DELETE /api/acr/{id}`

- 👤❌
- 🛠️ Admin → ❌ (avoid deleting evaluation data)

🧠 Additional API (IMPORTANT)

📊 Total Score

`GET /api/acr/summary/{facultyId}`

👉 Returns:

- totalScore (max 25)

PART A Score Summary

🧱 1. Data Type Mapping

👉 **No entity/table required**

API Endpoint

🔹 Base URL

`/api/part-a-summary`

✅ GET Summary

`GET /api/part-a-summary/{facultyId}`

📤 Response Should Contain:

- teachingScore (A)
- feedbackScore (B)
- deptActivityScore (C)
- universityActivityScore (D)
- socialScore (E)
- industryScore (F)
- acrScore (G)
- totalFacultyScore
- totalHodScore
- totalDirectorScore

🧠 Role-Based Access

Role	Access
👤 Faculty	Can view own summary
🛠️ Admin	Can view any faculty